

Nuclear Diffusion Theory in a CANDU Reactor

A first-year university guide to the physics blueprint of the CANDU refuelling game

SOURCE STATUS

This guide records bounded G4-R6/G5 PASS and G6 CONDITIONAL PASS evidence. The latest direct pinned headless run passed Core 158/158 and Golden 19/19; the Phase 6 focused run passed 46/46. It is not a direct external/full-core reactor simulator, and no applicable RRS comparison package or production authority has been admitted.

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Status: Living documentation; update when specifications, implementation, validation, or game rules change.

The short project answer

What physics is implemented now?

A bounded static-solver foundation is implemented in Core. It is not a feature-complete or validated reactor simulator. The completed Phase 3 work provides deterministic topology, inventory, snapshots, clock/queue, archive, replay, atomic application, and invalid-data contracts. G3 is a technical PASS.

The completed Phase 4 slices provide:

- P4-T01: deterministic topology-driven node, neighbour, and boundary-stencil assembly;
- P4-T02: coefficient/conductance binding and a matrix-free removal-plus-leakage operator; and
- P4-T03: one immutable two-group source/eigen iteration with deterministic scalar Jacobi inner solves, eigenvalue update, and power normalization; and
- P4-T04: caller-supplied outer convergence, zero-safe residual diagnostics, deterministic failure reasons, and fail-closed invalid-state results.
- P4-T05: synthetic one-node algebraic/no-leakage, symmetric, homogeneous, deliberate-nonconvergence, and canonical-order determinism coverage.
- P4-T08: a package-free synthetic Core benchmark that reports machine-specific timing and allocation observations without establishing a performance target.
- P4-T06-R3: a frozen boundary for an offline reduced/synthetic producer and interpolation data, with the existing coefficient-table contract as the only runtime-facing material-data surface.
- P5-T01: declarative 4-bundle/8-bundle scheme metadata, canonical insertion and discharge position plans, explicit flow-direction binding, and fail-closed catalog validation.
- P5-T02: an immutable, all-or-nothing location-layer shift that preserves retained bundle identities and represented basic state, creates fresh zero-energy entries, and returns deterministic ordered discharge states.
- P5-T03: named immutable refuelling mapping and atomicity event bodies, canonical position bindings, paired event-log append, and basic discharge audit records over the current Core bundle fields.
- P5-T04: immutable accepted-power projections, explicit positive-time left-endpoint energy integration, derived burnup, monotonicity/overflow checks, and proposed core-state-version advancement over the current Core bundle fields.
- P5-T05: immutable burnup-indexed coefficient values, strict ordered table metadata, coefficient-domain validation, exact-knot and linear-interpolation lookup, derived $\chi_{i,2}$, and fail-closed out-of-range behavior.
- P5-T06: explicit epoch/period/index spatial cadence, burnup-table-to-node coefficient rebinding, preserved validated conductances, deterministic spatial solving, and atomic lifecycle spatial/power binding acceptance.
- P5-T07: explicit Phase 5 invariant reports for bundle count, identity, location, nonnegative derived burnup, exact energy accounting, interval power-times-time accounting, monotonicity, and refuelling preservation, insertion, and discharge partition checks.
- P5-T08: test-only deterministic multistep S4/S8 history evidence with explicit interval/refuelling ordering, shuffled-input replay equality, per-bundle burnup records, ordered discharge traces, and identity non-reuse.
- P5-T09: complete I/Xe-bearing lifecycle, discharge, power-history, digest, and transaction contracts over the frozen Phase 2 state boundaries.
- P5-T10: structural sequence mapping for the G4-R6-approved ReducedModel authority, including exact final-node burnup mapping and the approved golden consumer.
- P5-T11 through P5-T16: complete-state burnup and lookup binding, source and history digest authentication, lifecycle-bound snapshot state, persistence- safe archive round-trip, restore-time reconciliation, and restart-self- contained snapshot reconstruction.

- P6-T01 through P6-T07: a limited, engine-neutral regulating-system contract for 14 logical liquid zones, adjusters, bulk poison, controller projections, queue transitions, and five deterministic synthetic scenarios. This is synthetic/test-only evidence, not an external reactor-control model.

These slices now establish a bounded, project-authored ReducedModel validation baseline for the engine-neutral Core. G4-R6 is PASS for the selected ReducedModel scope, and G5 is PASS for Phase 5 within that same boundary. The result includes approved golden-consumer evidence and the complete-state correction chain through P5-T16. It does not establish direct production or full-core CANDU/DONJON5/DRAGON5 authority, production thresholds, or a playable CLI game loop. R4/R5 and the earlier G4-R3/G4-R5 dispositions remain preserved as historical candidate/no-admission evidence; RRS and poison runtime behavior remain later-phase work. P5-T02 is bounded to the fields represented by the current immutable BundleInventory; fresh-template resolution, complete discharge/event envelopes, full power-history/digest lifecycle binding, coefficient lookup, and later physics integration were completed only by later bounded tasks. P5-T03 adds the approved mapping/atomicity body shapes and basic discharge evidence, but does not yet claim the complete I/Xe-bearing DischargeRecordV1 or RefuelTransactionV1 projection. P5-T04 integrates only the current energy/mass fields. P5-T05 adds the bounded coefficient-table domain and lookup contract, but does not claim canonical data-pack byte serialization/checksum verification, solver-side coefficient recomputation, the complete P2-T03 PowerSnapshot, or a single-owner lifecycle commit. P5-T06 adds a bounded coefficient-to-spatial- state recomputation boundary and exact periodic cadence/lifecycle binding, but does not claim the full scheduler, kinetics/I-Xe transitions, canonical data-pack checksum serialization, complete power-history binding, or reference and golden evidence. P5-T07 adds the exact bounded invariant checks over the current Core inventory, burnup interval, and refuelling result contracts. P5-T08 adds deterministic synthetic multistep history evidence by composing those contracts, but does not claim a production history schema, complete I/Xe/lifecycle/discharge envelope, reference comparison, or golden evidence. P5-T09 now supplies the complete bounded lifecycle/I-Xe/power-history and transaction contracts. P5-T10 is complete for the G4-R6-approved ReducedModel path and supplies structural sequence evidence; P5-T11 through P5-T16 complete the authenticated transition, persistence, and restart boundary. Phase 5/G5 is closed PASS for the approved ReducedModel/Core scope. Phase 6/G6 is a **CONDITIONAL PASS** for the bounded synthetic/test-only Core/ReducedModel scope: the implementation is complete for that evidence boundary, but an applicable RRS comparison package is still required before an unconditional or external claim is possible.

The 2026-08-24 recovery handoff also completed TEST - INFRA -01, P6 - INTEGRATION -01, P6 - INTERNALS -01, and CORE - NAMING -01. Those tasks refreshed test artifact portability, the RRS queue-admission seam, internal canonical-byte ownership, and the task-ID compatibility inventory. They did not change the physics model, public schema, golden authority, or G6 scope. STATUS-VISIBILITY -01 is complete as the derived-documentation handoff; no named next task is selected until the owner decides whether to start planned Phase 7 or re-sequence the explicitly non-physics Phase 8 CLI slice. The project scope register and linked reports remain authoritative.

The implemented Phase 1 work is an **offline reference pipeline**:

- pinned DRAGON5 and DONJON5 smoke-case provenance and runners;
- strict parsers/exporters for selected reference output;
- compact parser-only fixtures, manifests, hashes, and repeatability checks.

Those tools establish how future evidence may be collected. They do not run at game time and their KINF or K-effective observations are not runtime constants, solver targets, or golden data.

What exists as a physics design?

The completed P2-T01 through P2-T05 tasks define frozen, engine-neutral implementation inputs for:

- explicit CANDU-style topology and boundary metadata;
- a static two-group diffusion eigenproblem;
- power normalization and deterministic source iteration;
- refuelling, bundle identity, burnup, and coefficient interpolation;
- point kinetics, I-135/Xe-135, regulating control, and optional feedback;

- observables, deterministic serialization, and validation evidence.

G2 is FORCED CLOSED / WAIVED: that administrative decision permits use of the existing specifications where no critical blocker is introduced, but it is not a technical PASS and does not approve a new equation, tolerance, golden value, or public contract. The specifications remain the blueprint for the remaining implementation; their numerical acceptance thresholds and production reference comparisons remain owned by later gates.

What remains before the static solver is feature-complete and validated?

1. G4-R6 has approved the bounded project-authored ReducedModel authority and six quantity-specific profiles. The earlier G4-R3/G4-R5 blockers remain historical records and do not become production approval.
2. G5 is complete for the approved ReducedModel/engine-neutral Core scope. P5-T10 through P5-T16 provide structural sequence, complete-state, authentication, persistence, and restart evidence; the final validation snapshot is recorded in Section 6.7 and the G5 gate report.
3. Direct production/full-core external authority and production thresholds remain deferred. G6 conditionally passes the synthetic/test-only regulating system; an applicable RRS comparison package and G6 re-entry remain needed. Phase 7A adds kinetics and xenon; Phase 7B remains optional feedback work.

1. Why a reactor can become a game

A reactor is a system in which neutrons cause fissions, fissions release heat, and the player changes the conditions that determine how many neutrons survive to cause more fissions. The game does not need to reproduce every detail of a power station. It needs a stable chain of cause and effect:

```
player action
-> fuel/control state changes
-> neutron distribution and power change
-> burnup and isotope history change
-> later choices become easier or harder
```

The intended game is a deterministic, text-first strategy game about online refuelling of a CANDU-style reactor. The player chooses when and where to move fuel and how to use limited regulating devices. The simulation then explains the consequences through power, flux shape, burnup, xenon, tilt, and control state.

This is an entertainment simulation, not operator training, a safety analysis, or a model of a named commercial station.

2. Background: atoms, fission, and chain reactions

2.1 The nucleus and isotopes

An atom has a small nucleus surrounded by electrons. The nucleus contains protons and neutrons. An isotope is identified by its number of protons and neutrons. Uranium-235, for example, is an isotope that can fission after absorbing a neutron.

In fission, a heavy nucleus absorbs a neutron and splits into lighter nuclei. The event releases:

- kinetic energy of the fission fragments, which becomes heat;
- additional neutrons;
- radioactive fission products; and
- gamma radiation and other energy.

Some emitted neutrons are fast. They lose energy by colliding with matter. A slow neutron is called thermal when its energy is close to the thermal motion of the surrounding material. Thermal neutrons are especially important in a heavy-water moderated reactor because they are effective at causing fission in the fuel.

2.2 The chain reaction

The number of neutrons in one generation compared with the previous generation is summarized by the effective multiplication factor, k :

```
k > 1 -> neutron population tends to grow
k = 1 -> neutron population is steady
k < 1 -> neutron population tends to shrink
```

k is dimensionless. It is not the same as power. A reactor can have a stable shape calculation with a particular k , while a separate amplitude variable sets the actual power represented by that shape.

The project uses reactivity as a convenient signed quantity:

$$\rho = (k - 1) / k$$

Positive ρ tends to increase power amplitude; negative ρ tends to decrease it. The project does not add separate hidden ρ_{xenon} or $\rho_{\text{controller}}$ terms. Those effects change absorption coefficients before the spatial solve, and the resulting k supplies the one reactivity value.

3. What makes a CANDU-style reactor special?

The guide uses CANDU-style language without claiming a specific station geometry. The important concepts are:

- **Heavy water:** deuterium oxide is used as moderator and, in a typical CANDU arrangement, coolant. Its low neutron absorption helps make efficient use of natural uranium fuel.
- **Pressure-tube channels:** fuel is arranged in many horizontal channels rather than one large pressure vessel. The project represents a production configuration as 380 channels with 12 bundle positions per channel, giving 4,560 explicit channel-position nodes.
- **Fuel bundles:** each position owns a persistent bundle identity. A bundle can move, accumulate energy and burnup, carry I/Xe history, and eventually be discharged.
- **On-power refuelling:** fuel movement can occur while the reactor is operating. The game turns this into a central decision: refuelling changes local reactivity and future fuel value without resetting the whole core.
- **Regulating devices:** liquid-zone controllers, adjusters, and moderator poison can change local absorption and therefore power shape. The project includes only limited regulating behavior, not shutdown or accident systems.

The literature digest contains useful CANDU context, including multigroup to two-group workflows, burnup/refuelling histories, xenon, and regulating-device roles. It does not authorize a runtime number or a golden case. Relevant rows include S1-R02, S1-R03, S1-R05, S1-R06, S1-R07, S1-R08, S1-R09, S5-R03, S5-R04, S5-R08, S6-R02, S6-R03, and S6-R05.

4. From neutron transport to diffusion theory

4.1 The transport problem

The most detailed common description follows neutrons in position, direction, energy, and time. It asks questions such as:

- where is a neutron?
- which direction is it moving?
- what is its energy?
- will it scatter, be absorbed, or cause fission?

That full transport description is expensive. A game needs a reduced model that preserves important spatial and temporal cause-and-effect at interactive speed.

4.2 The diffusion approximation

Diffusion theory makes a simplifying assumption: in a sufficiently scattering, nearly isotropic region, neutron flow can be estimated from the gradient of neutron flux. In ordinary notation, a one-group diffusion balance resembles:

$$-\text{div}(D \text{ grad}(\phi)) + \text{Sigma_a} * \phi = \text{source}$$

Here ϕ is scalar flux, D is a diffusion coefficient, and Sigma_a is a macroscopic absorption cross section. The first term represents leakage from one region to another; the second represents neutrons removed by absorption.

The project does **not** derive its runtime spatial coefficients from a guessed geometry formula. The candidate P2-T02 contract uses preassembled finite-volume edge conductances T and boundary conductances B . This keeps the topology, units, and coefficient provenance explicit.

Diffusion is an approximation. It becomes less reliable near sharp voids, strongly directional streaming, or detailed boundaries. Those limitations are part of why DRAGON5/DONJON5 remain offline reference tools rather than runtime dependencies.

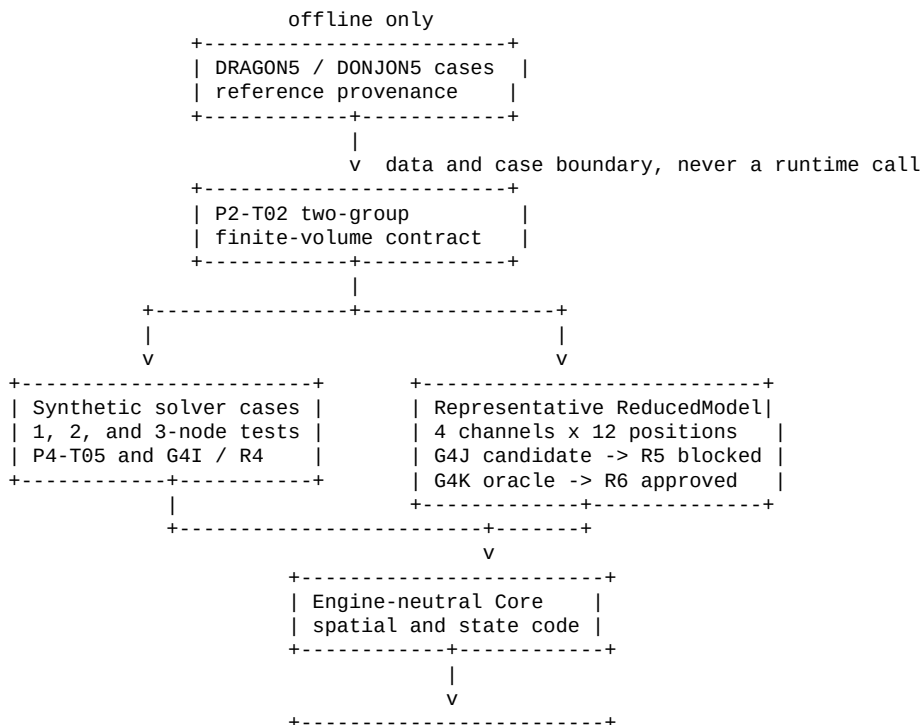
4.2.1 The reduced offline data boundary

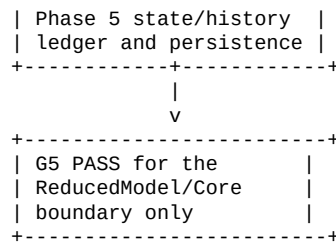
The game-facing model is intentionally smaller than a transport calculation. P4-T06-R2D retired direct Candu6.x2m admission into the P2-T02 runtime contracts, and P4-T06-R3 freezes the replacement boundary. An offline synthetic or separately admitted reduced producer may create material values for burnup-indexed interpolation, but the runtime consumes only the existing validated `BurnupCoefficientTableV1` projection.

The table owns material coefficients, burnup knots, material identity, units, provenance, version, and checksum identity. It does not own topology, flow direction, node volume, edge or boundary conductance, flux, power, normalization, or convergence policy. Those values remain explicit inputs to the existing topology-bound `SpatialRecomputeRequestV1` and P2-T02 solver. Runtime lookup is exact-knot or linear interpolation within the table domain; out-of-range, invalid, stale, or ambiguously mapped data fails closed. The reduced producer, reference tools, exact reference data, and raw artifacts remain offline. See the R3 boundary specification.

4.2.2 Models used by tests and gates

The project uses several related models for different kinds of evidence. They must not be read as one interchangeable reactor model. A test can pass because it proves a contract, a gate can pass for a bounded `ReducedModel` domain, and the direct production or full-core authority can still remain deferred.





The model cards below explain what each layer means.

- **Offline reference model.** DRAGON5 and DONJON5 are used to establish reference-case provenance and, when separately admitted, compact offline data. They are never called by the runtime. No current G4 or G5 result claims direct DRAGON5, DONJON5, or named-station authority.

- **Frozen two-group diffusion contract.** P2-T02 defines the finite-volume two-group eigenproblem, conductance-based leakage, source ordering, signs, normalization, and convergence diagnostics. P4-T01 through P4-T05 exercise this contract in engine-neutral Core code. It is the approved equation and data boundary, not by itself a full-core validation result.

- **Synthetic solver model.** P4-T05 uses the one-node/no-leakage algebraic example, a two-node reciprocal symmetric case, and a homogeneous three-node chain, plus deliberate nonconvergence and input-order permutations. P4-T06-G4I adds a heterogeneous four-node manufactured solution. These models test arithmetic, signs, normalization, failure reporting, and determinism. They are synthetic mathematical checks and cannot become direct CANDU authority.

- **Representative ReducedModel authority.** The G4K package retains the explicit 4-channel by 12-position topology: 48 nodes, 92 reciprocal edges, 104 boundaries, alternating flow, burnup projection, two S4 shift directions, and overlay cases. The five scenarios are `fresh_start`, `equilibrium_like`, `refuelled_4_bundle_shift`, `rrs_tilt_perturbation`, and `bulk_poison_perturbation`. G4J was a deterministic project-authored candidate, but G4-R5 correctly left it Candidate / Deferred / NoGolden. G4K generated expected values through a separate Python/NumPy dense generalized-eigen path. G4-R6 then approved an `ApprovedGolden` payload and six quantity-specific profiles for the bounded `ReducedModel` domain only.

- **Phase 5 state/history model.** P5-T01 through P5-T16 add the ledger around the spatial model: explicit S4/S8 movement, bundle identity and location, accepted power, energy and burnup, coefficient lookup, spatial recompute cadence, I/Xe-bearing lifecycle state, event history, digest binding, and save/load restoration. P5-T10 and G5 use the approved `ReducedModel` projection to check structural sequence mapping and exact accounting/authentication contracts. They do not claim an independent external numerical-drift result.

- **RRS and feedback extension.** The RRS tilt and bulk-poison cases above are controlled overlays in the bounded `ReducedModel` evidence. P6-T01 through P6-T07 now implement the limited 14-zone, adjuster, poison, controller, queue, and scenario contracts as synthetic/test-only Core evidence. G6 is a `CONDITIONAL PASS`, because no applicable external or production RRS comparison package is admitted.

Evidence	Model used	What a passing result means	Boundary that remains
P4-T05 synthetic tests	Algebraic one-node and toy two/three-node diffusion cases	Frozen operator, normalization, diagnostics, nonconvergence, and deterministic ordering behave as specified	No external or production numerical authority
P4-T06-G4I / G4-R4	Four-node heterogeneous manufactured solution	Synthetic spatial behavior can be reproduced and, for the bounded synthetic domain, conditionally admitted	Not a CANDU, transport, burnup, or full-core result
P4-T06-G4J / G4-R5	Project-authored 48-node <code>ReducedModel</code> candidate with five scenarios	Candidate generation and Core comparison are deterministic	Candidate remained unapproved; G4-R5 remained blocked
P4-T06-G4K / G4-R6	Same bounded <code>ReducedModel</code> domain with independent dense generalized-eigen expected values	Five scenarios and six quantity profiles support <code>ApprovedGolden</code> use for <code>ReducedModel</code>	No direct DRAGON5/DONJON5, production, release, safety, or 380-channel claim
P5-T10 / G5	<code>ReducedModel</code> sequence plus the Phase 5 bundle/lifecycle ledger	Refuelling movement, identity, energy/burnup mapping, digests, persistence, and restart contracts pass exactly within the admitted scope	Not an independent external burnup or full-core numerical comparison

Evidence	Model used	What a passing result means	Boundary that remains
Phase 6 / G6	Synthetic/test-only RRS state, actuator, queue, and five-scenario model	P6-T01 through P6-T07 complete; G6 CONDITIONAL PASS; 46 focused tests and the full regression pass	No applicable RRS comparison, production, external, or golden authority

This map is the safest way to interpret the evidence: G4-R5 did not fail to clear a model that G4-R6 later silently relabelled. R5 rejected the candidate as authority. G4K supplied a new independent authority package, and R6 passed only the narrower ReducedModel scope that package actually covered.

4.3 Homogenization and energy groups

A real neutron spectrum is continuous in energy. A group model divides it into energy ranges and replaces detailed behavior by group-averaged coefficients. The project uses two groups:

- **Group 1:** fast neutrons;
- **Group 2:** thermal neutrons.

The first specification allows downscatter from group 1 to group 2 and rejects upscatter. Each node/group stores or receives quantities such as:

Quantity	Meaning	Unit
phi_g	scalar neutron flux	$m^{-2} s^{-1}$
Sigma_a,g	total absorption	m^{-1}
Sigma_f,g	fission part of absorption	m^{-1}
nu*Sigma_f,g	fission cross section weighted by neutron yield	m^{-1}
Sigma_s,1->2	fast-to-thermal downscatter	m^{-1}
chi_g	fission spectrum fraction	1
V	node volume	m^3
E_f	energy per fission used for game power	J

The invariant $\text{Sigma}_f \leq \text{Sigma}_a$ prevents fission absorption from being counted twice. The fission spectrum fractions sum to one. All authoritative arithmetic uses finite double values.

5. The static two-group model: specified and bounded implementation

P4-T01 through P4-T05 implement the topology/stencil, coefficient/operator, one-iteration, outer-diagnostics, and synthetic-case portions of the approved P2-T02 model. This section explains the complete specified target. A returned synthetic P4-T04 state is not a direct external reference result. The selected ReducedModel authority is separately admitted by G4-R6 for its bounded scope; that admission does not claim full-core or production CANDU validation.

5.1 Nodes and topology

The spatial node is the explicit pair:

```
node = (ChannelId, BundlePosition)
```

Channel IDs, positions, coordinates, neighbours, flow direction, and boundary faces are explicit records. Array order is only a deterministic evaluation order; it never means physical adjacency or coolant direction.

For a production-sized configuration:

```
380 channels * 12 positions = 4,560 nodes
```

The topology contract distinguishes transverse channel neighbours from within-channel neighbours and requires reciprocal records. Boundary faces are labelled Reflective, Vacuum, or SpecifiedLeakage. A missing boundary conductance is invalid rather than silently treated as a default.

5.2 Two-group eigenvalue equations

Define removal coefficients for the two groups:

$$\begin{aligned}\text{Sigma}_{r,1} &= \text{Sigma}_{a,1} + \text{Sigma}_{s,1 \rightarrow 2} \\ \text{Sigma}_{r,2} &= \text{Sigma}_{a,2}\end{aligned}$$

The specified steady equations are:

$$\begin{aligned}A_1 \phi_1 &= (\chi_1 / k) * F(\phi) \\ A_2 \phi_2 &= \text{Sigma}_{s,1 \rightarrow 2} * \phi_1 + (\chi_2 / k) * F(\phi) \\ F_i(\phi) &= \nu * \text{Sigma}_{f,1,i} * \phi_{1,i} \\ &\quad + \nu * \text{Sigma}_{f,2,i} * \phi_{2,i}\end{aligned}$$

A_g combines removal and leakage. The positive fission and downscatter terms are sources on the right-hand side. k is the eigenvalue that makes the neutron balance self-consistent.

5.3 Finite-volume leakage

For node i and group g , the candidate finite-volume operator is:

$$\begin{aligned}(A_g \phi_g)_i &= \text{Sigma}_{r,g,i} * \phi_{g,i} \\ &\quad + (1 / V_i) * [\\ &\quad \quad \text{sum}_j T_{g,ij} * (\phi_{g,i} - \phi_{g,j}) \\ &\quad \quad + \text{sum}_f B_{g,if} * \phi_{g,i} \\ &\quad]\end{aligned}$$

The interior term is conservative: the same conductance appears on both sides of a reciprocal edge, with opposite flux differences. A high-flux node loses neutrons through the left-hand leakage term; leakage is not a source.

Reflective faces use $B = 0$. Vacuum and specified-leakage faces require a positive, approved boundary conductance. The specification does not invent a Marshak factor or derive a coefficient from an undocumented geometry.

5.4 Power and normalization

Local integrated fission power is specified as:

$$\begin{aligned}P_i &= V_i * E_{f,i} * [\\ &\quad \text{Sigma}_{f,1,i} * \phi_{1,i} \\ &\quad + \text{Sigma}_{f,2,i} * \phi_{2,i} \\ &\quad] \\ P_{\text{total}} &= \text{sum}_i P_i\end{aligned}$$

The caller supplies a positive target power P_{target} in watts. After a valid nonnegative shape is found:

$$\begin{aligned}\alpha &= P_{\text{target}} / P_{\text{total}} \\ \phi_{g,i} &= \alpha * \phi_{g,i}\end{aligned}$$

This changes flux amplitude and power, not k or the relative shape. The solver records power before/after normalization, α , and power-balance error. Power normalization is separate from criticality.

5.5 One implemented iteration and the outer loop

P4-T03 implements one deterministic source/power iteration of this sequence:

- Step 1: validate topology, coefficients, target power, initial flux, and policy.
- Step 2: normalize the initial nonnegative shape.
- Step 3: compute fission source F and production total Q .
- Step 4: solve the group 1 linear system.
- Step 5: solve the group 2 system using downscatter plus fission source.
- Step 6: reject failed, negative, or non-finite trial flux.
- Step 7: update k from the ratio of new to old production.

- Step 8: normalize the trial shape again and recompute F and Q from the new normalized state, never stale trial values.
- Step 9: in P4-T04, test convergence or continue with the normalized state.

P4-T03 returns after the normalized step. P4-T04 places repeated steps in the specified outer convergence decision and exposes the required diagnostics. The implemented inner method is deterministic scalar Jacobi under a caller-supplied policy; parallel reduction, Burst, lower precision, and unordered sums are not part of v1.

5.6 Specified convergence and implemented outer diagnostics

The P2-T02 outer-solve contract records:

```
delta_k_abs
delta_k_rel
residual_absolute_inf
residual_relative_inf
source_shape_change_inf
power_balance_rel
```

Convergence requires all four conditions to pass: an absolute or relative k change condition, residual condition, source-shape condition, and power-balance condition. The caller supplies every tolerance and maximum iteration count; the solver must not invent or relax them. P4-T04 implements this decision and does not claim a new numeric threshold.

The complete result must fail closed on NaN, infinity, negative flux, zero production, invalid conductance, failed inner solve, rejected upscatter, attempted clamp, or nonconvergence. It must not return the last iterate as a successful solve.

5.7 Synthetic case coverage

P4-T05 exercises the frozen contract with committed synthetic inputs only:

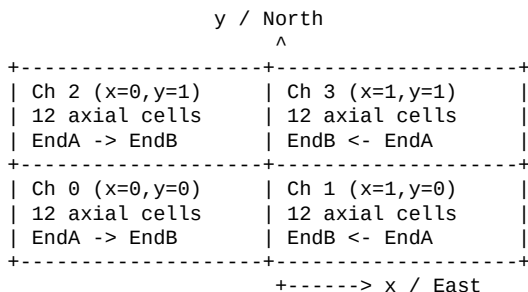
- the P2-T02 one-node/no-leakage example is checked algebraically, including $k=0.6875$ and the normalized shape ratio;
- the smallest valid runtime topology has two bundle positions under P2-T01, so a two-node reciprocal case checks equal-flux symmetry and normalization;
- a homogeneous three-node chain checks nodewise shape preservation;
- a one-iteration policy exhaustion checks deterministic nonconvergence, final diagnostics, and the absence of a usable state; and
- reversing input coefficient record order must produce identical diagnostics and final flux arrays.

These checks establish synthetic contract coverage only. They do not compare against DONJON5, create golden data, or approve numerical tolerances.

5.8 Model geometry and ReducedModel visualizations

The approved geometry has two useful views. The channel-plane view shows the logical 2x2 placement and the explicit transverse links. The elevation view below it shows the twelve axial control volumes inside each channel. The integer x and y coordinates are topology coordinates, not inferred metre distances; physical lengths and node volumes remain explicit SI data.

Channel-plane cross-section (logical x/y; each box contains 12 axial cells)



transverse links: 0-1 East/West, 0-2 North/South,
1-3 North/South, 2-3 East/West

This geometry view is a visualization of the admitted ReducedModel input, not a claim that a rectangular formula can replace the topology table. Every channel and every exposed face is still represented by explicit records. The four channel-plane links are repeated at each of the twelve positions, while the two channel ends add the explicit EndA and EndB faces for every channel. The resulting case has 48 spatial nodes, 92 reciprocal edges, and 104 boundary records.

The elevation/topology view makes the axial indexing and alternating flow visible:

The approved G4-R6 case is a compact spatial model, not a miniature 380-channel plant model. Its topology is explicit and its array order is only an evaluation order. The following schematic shows the shape of the case without implying that a row in memory is a physical flow direction.

```
48 spatial nodes = 4 channels x 12 axial bundle positions

channel 0  (0,0) -- (0,1) -- ... -- (0,11)      flow ->
channel 1  (1,0) -- (1,1) -- ... -- (1,11)      flow <-
channel 2  (2,0) -- (2,1) -- ... -- (2,11)      flow ->
channel 3  (3,0) -- (3,1) -- ... -- (3,11)      flow <-
           ^       ^       ^
           +-----+-----+ explicit reciprocal transverse edges
```

Each position also has explicit boundary-face records and two energy groups.
The authority package binds 92 reciprocal edges and 104 boundaries.

At each node, the local material state is selected first and the spatial solve then couples it to neighbouring nodes:

```
burnup B_b
  |
  v
coefficient table lookup/interpolation
  |
  v
node i = (channel, position)
  |
  +--> fast flux phi_1 ---- downscatter ----> thermal flux phi_2
  |
  +--> absorption and fission source, with k
  |
  +--> reciprocal edge leakage T and boundary leakage B
  |
  v
normalized flux shape, node power, total power, residuals, convergence
```

The Phase 5 history model wraps that spatial solve in an explicit ledger. This is the model exercised by P5-T10 and the final G5 correction chain:

```
accepted bundle power P_b(t_n), time step Delta_t
  |
  v
cumulative energy += P_b(t_n) * Delta_t
  |
  v
burnup = initial burnup + energy / heavy-metal mass
  |
  v
bind the burnup-indexed coefficient result
  |
  v
recompute affected spatial state at its explicit cadence
  |
  v
append history, authenticate digests, save or restore fail-closed
```

G5 passes when this ledger preserves the frozen identities, event order, complete histories, lookup bindings, and authenticated persistence within the approved ReducedModel/Core boundary. It does not turn the ledger into a direct external reactor reference.

6. Fuel movement, burnup, and coefficient tables

6.1 Bundle energy and burnup

For bundle b , the authoritative state is cumulative fission energy and heavy metal mass. Current burnup is derived:

$$B_b = \text{InitialBurnup}_b + \text{CumulativeFissionEnergy}_b / m_{\text{HM},b}$$

For a left-endpoint interval with accepted power $P_b(t_n)$:

$$\begin{aligned} \Delta E_b &= P_b(t_n) * \Delta t \\ \text{CumulativeFissionEnergy}_{\text{next}} &= \text{CumulativeFissionEnergy} + \Delta E_b \\ B_{\text{next}} &= \text{InitialBurnup} + \text{CumulativeFissionEnergy}_{\text{next}} / m_{\text{HM}} \end{aligned}$$

The units are explicit:

$$\begin{aligned} W * s &= J \\ J / \text{kg}_{\text{HM}} &= J/\text{kg}_{\text{HM}} \end{aligned}$$

The interval is split at every event. Burnup is monotone and no step reduction, clamp, or extrapolation is silently applied.

P5-T04 implements this rule as an immutable `Core` transition over the current `BundleInventory` representation. The accepted power projection must bind to the exact current time, core-state version, and opaque source-state digest, and contain one finite, nonnegative sample for every live bundle at its explicit node. Samples and result records use canonical (`ChannelId`, `BundlePosition`, `BundleId`) order. The transition computes every proposed energy and burnup value before constructing a replacement inventory, increments the proposed core-state version exactly once, and marks the consumed power snapshot invalid. Missing, stale, non-finite, overflowing, or non-monotone values reject with the source inventory unchanged. The complete P2-T03 snapshot digests/history, coefficient lookup, and lifecycle-owner commit remain later bounded work.

6.2 Declarative refuelling

V1 defines complete 4-bundle and 8-bundle schemes, called S4 and S8. A scheme is one atomic transaction:

- inserted bundles receive explicit templates and fresh identities;
- bundles shift toward the declared flow endpoint;
- the boundary train is discharged;
- existing bundle identities, energy, burnup, and I/Xe history move with them;
- coefficients are looked up for the proposed state; and
- the whole operation commits only if every check succeeds.

`EndAtoEndB` binds to `TowardEndB`; `EndBtoEndA` binds to `TowardEndA`. Equal-time commands form one all-or-none batch. A failed precondition or postcondition restores the complete pre-state byte-for-byte.

P5-T01 implements the declarative scheme boundary in `Core`. A validated catalog must contain both four-bundle and eight-bundle coverage; each definition keeps the inserted positions in ascending physical order and produces the endpoint discharge positions for the channel's explicit flow direction.

P5-T02 applies one of those plans as a pure immutable replacement. The target channel must be fully occupied; retained bundles move by explicit position mapping while keeping their stable IDs and represented basic fields, inserted states arrive at the plan's positions with zero cumulative energy, and the discharged states are returned in ascending old-position order. Any failed precondition leaves the original inventory unchanged. This slice does not yet resolve complete fresh-fuel templates, carry the later I/Xe/lifecycle envelope, append a command/event/discharge record, integrate burnup, or look up coefficients.

P5-T03 adds the approved `RefuelMappingBodyV1` and `RefuelAtomicityBodyV1` shapes. It validates explicit applicability tags, canonical one-channel position coverage, paired event-envelope time/status, and appends the mapping/atomicity pair to an immutable event log only when both records succeed. It also returns deterministic basic discharge audit records containing the fields currently represented by `BundleState`. P5-T03 itself did not claim the complete I/Xe-bearing discharge and transaction projections; P5-T09 later completes those bounded contracts. No digest is invented by the P5-T03 slice.

6.3 Burnup-indexed interpolation

Each material variant resolves to a versioned table with strictly increasing burnup knots. For a value between adjacent knots:

$$\begin{aligned}\alpha &= (B - B_j) / (B_{(j+1)} - B_j) \\ c(B) &= (1 - \alpha) * c_j + \alpha * c_{(j+1)}\end{aligned}$$

The canonical χ_1 is interpolated and χ_2 is derived as $1 - \chi_1$. Values outside the table domain are rejected. The implementation must not extrapolate, clamp, repair an invalid coefficient, or independently interpolate the derived spectrum component.

P5-T05 implements this bounded contract with immutable `BurnupCoefficientValuesV1`, `BurnupCoefficientRowV1`, `BurnupCoefficientTableV1`, and lookup-result records. Table identity, schema and data versions, material/unit identity, path-free provenance, and an opaque loader-supplied checksum are retained with the lookup. Rows must be supplied in strict increasing burnup order; finite coefficient relationships are validated before storage and again after interpolation. Exact knots select a single row, interior values use the adjacent bracket, and the derived spectrum component is never independently interpolated. Missing metadata, invalid coefficients, reordered/duplicate knots, private file-path provenance, and out-of-range burnup fail closed. Canonical table-byte serialization and checksum verification remain a data-pack boundary rather than an invented runtime serializer in this slice.

6.4 Affected coefficient and spatial-state recomputation

P5-T06 adds the narrow boundary between the burnup table and the existing two-group spatial solver. A cadence stores an explicit nonnegative epoch, a strictly positive period, and an integer event index. The next event is formed from $\text{epoch} + \text{event_index} * \text{period}$ using the stored binary64 values. A result that is not finite, representable, or strictly later when the index advances fails closed; wall-clock time and Unity frame timing do not participate.

At the scheduled time, every live spatial node resolves the current bundle's derived burnup against the table for its material variant. The request also supplies one explicit positive node volume per node. The lookup result records the bundle, node, material, burnup, table identity, bracket, and interpolation fraction that supplied the node's material coefficients. The recomputation does not change the approved table rule: exact knots select one row, interior values use the adjacent linear bracket, and out-of-range values are rejected.

The existing validated edge and boundary conductances are copied in canonical topology order. They are not re-derived from a guessed geometry or silently replaced while burnup changes the node coefficients. The existing P2-T02 solver and its caller-supplied inner and outer policies then calculate the replacement spatial state and normalize it to the requested positive target power. No hidden tolerance or convergence rule is introduced by P5-T06.

Lifecycle acceptance is the final operation. The request binds exact simulation time, core/bundle/topology/data-pack versions, topology/data/state digests, and explicit spatial-solve and power-snapshot identities. The conductance stencil must come from the exact lifecycle topology instance, not merely a same-sized topology. Only a converged solve with a valid next cadence event can produce a replacement lifecycle; failed lookup, invalid state, missing conductance, nonconvergence, stale identity, or cadence failure produces no accepted replacement. The contract is intentionally not the full P2-T04 scheduler, kinetics, I/Xe, or regulating-system transition, and it does not serialize or verify data-pack checksum bytes.

6.5 Phase 5 identity, location, burnup, and energy invariants

P5-T07 adds a fail-closed invariant boundary over the bounded Phase 5 contracts. The expected live bundle count is an explicit caller input; it is not inferred from a partial inventory or silently replaced with a production default. The validator checks the exact count, globally unique stable bundle identities, unique legal (`ChannelId`, `BundlePosition`) locations, finite nonnegative initial burnup and cumulative fission energy, positive heavy-metal mass, and the exact derived relation $B = \text{InitialBurnup} + \text{CumulativeFissionEnergy} / m_{\text{HM}}$.

For a burnup interval, the validator binds canonical (`ChannelId`, `BundlePosition`, `BundleId`) record order to the source and result inventories and to the accepted power snapshot. It checks exact binary64 $\Delta E = P * \Delta t$, exact cumulative-energy addition, derived burnup accounting, nonnegative monotonic burnup, explicit time/version binding, and the no-representable-increase failure condition. No tolerance, clamp, hidden step reduction, or unordered sum is introduced.

For a refuelling shift, the validator checks the explicit flow-to-direction binding, ordered discharged boundary states, exact retained identity and basic state preservation at the shifted locations, fresh zero-energy identities at the insertion positions, and the partition of retained, inserted, and discharged identities. This remains a bounded identity/location/energy check; complete I/Xe, power-history, residence, lifecycle, and long deterministic history envelopes remain later Phase 5 work and G5 evidence.

6.6 Deterministic multistep history evidence

P5-T08 exercises the approved immutable transitions as a bounded evidence harness. Each synthetic history accepts one explicit left-endpoint burnup interval, advances the synthetic core version once, then applies the equal-time S4 or S8 refuelling batch and advances the version once more. The interval is therefore split at the refuelling boundary: a fresh bundle gets no energy from the preceding interval and can contribute only to the next one.

The evidence covers repeated S4 cycles in EndA to EndB flow and repeated S8 cycles in EndB to EndA flow. At every step it checks exact count, location, energy, derived burnup, monotonicity, movement, insertion, discharge, and identity-lifetime invariants. Replaying the same history with accepted-power records supplied in reverse order produces the same complete trace, including every canonical per-bundle burnup record, every ordered discharged state, and the resulting live inventory fingerprint.

The trace fingerprint is test-only repeatability evidence. It is not a new runtime digest algorithm or a production serialization format. Direct Candu6.x2m admission was retired by R2D; the R4/R5 candidate pack and representative candidate remain preserved as Candidate/Deferred. G4-R6 admits a separate project-authored ReducedModel authority with six approved profiles, and G5 is PASS for the bounded Phase 5/Core sequence and lifecycle scope. Neither gate establishes direct production/full-core CANDU authority.

6.7 Phase 5/G5 status and validation snapshot

The final same-context G5 review returned PASS for the approved ReducedModel / engine-neutral Core scope. The result is a bounded validation and lifecycle approval, not a direct production or full-core external solver baseline.

Evidence	Result
P3-T04 serialization regression	5/5 PASS
P5-T11/P5-T16 focused complete-state regression	7/7 PASS
Phase 5 focused Core tests	44/44 PASS
Full Core and Golden suites	Core 110/110; Golden 19/19 PASS
Complete solution regression	129/129 PASS
G4-R6 approved consumer	3/3 PASS
P5-T10 sequence consumer	2/2 PASS
Profile digest audit and formatter	6 profiles PASS; clean

The review receipt is verified as `gpt-5.6-sol` with high effort in the reused reviewer context. The approved G4-R6 artifact and manifest hashes are unchanged. Direct production/external authority remains deferred, and the Phase 6/G6 result is recorded separately below.

6.8 Current test results and evidence boundary

The following checks were recorded or rerun on 2026-08-24 with the repository-pinned .NET SDK 10.0.302. They are implementation evidence, not a claim of production or external CANDU validation.

Check	Fresh result
Direct full headless solution regression	Core 158/158 PASS; Golden 19/19 PASS; zero failures and zero skips
Direct Phase 6 focused regression	Core 46/46 PASS; zero failures and zero skips
TEST-INFRA-01 artifact-output wrapper recovery	Core 156/156 and Golden 19/19 PASS; zero failures and zero skips

The direct run shows 177 passing headless tests in the normal repository layout. TEST-INFRA-01 closed the former artifact-output location defect for the recovery baseline; the later direct count includes the separately characterized Phase 6 integration and internal tests. All Phase 6 results remain synthetic/test-only contract evidence. G6 is still a CONDITIONAL PASS: no applicable RRS comparison package, production/full-core CANDU authority, or unconditional external claim has been admitted.

7. Point kinetics and xenon history

7.1 Quasi-static amplitude

The spatial solve supplies a shape normalized to a reference power P_{ref} . Point kinetics adds an amplitude a :

```
phi_actual,g,i = a * phi_shape,g,i
P_i            = a * P_shape,i
P_total        = a * P_ref
```

The amplitude changes power magnitude while the spatial solve determines shape. Every actual bundle power used for burnup must be bound to the amplitude, spatial solve, state versions, node volume, and coefficient digest.

7.2 Delayed-neutron point kinetics

The planned delayed-neutron groups use prompt generation time Λ , delayed fractions β_j , decay constants λ_j , amplitude a , and precursor states C_j :

```
da/dt = ((rho - beta) / Lambda) * a
        + sum_j lambda_j * C_j

dC_j/dt = (beta_j / Lambda) * a - lambda_j * C_j

beta = sum_j beta_j
```

For v1, the numerical update is explicit left-endpoint Euler:

```
a_next = a + Delta_t * da/dt
C_next,j = C_j + Delta_t * dC_j/dt
```

Negative or non-finite values fail closed. A future approved task may compare a different integrator, but it cannot replace Euler silently.

7.3 I-135 and Xe-135

The project attaches isotope state to the persistent bundle identity. There is no cross-bundle xenon transport or thermal-hydraulic mixing model in v1.

Using actual flux and fission-rate density:

```
R_f = sum_g Sigma_f,g * phi_actual,g

dN_I/dt = gamma_I * R_f - lambda_I * N_I

dN_Xe/dt = gamma_X * R_f + lambda_I * N_I
          - [lambda_X + sum_g sigma_Xe,g^a * phi_actual,g] * N_Xe
```

The neutron-absorption contribution is added to the spatial absorption:

```
Sigma_Xe,g^a = sigma_Xe,g^a * N_Xe
Sigma_a,g,eff = Sigma_a,g,base + DeltaSigma_a,g + Sigma_Xe,g^a
```

The base table must be explicitly xenon-free so dynamic Xe is not counted twice. I/Xe inventories are authoritative; number densities are derived from the current node volume. The next spatial solve is mandatory after Xe changes.

8. Regulating system and optional feedback

8.1 Total-power and tilt control

The limited RRS uses total-power error and regional tilt error:

```

PowerError = PowerSetpoint - MeasuredPower
P_region,r = sum_i_in_region_r P_i
f_region,r = P_region,r / P_total
TiltError_r = TargetFraction_r - f_region,r

```

The planned controller is PI-like, with optional regional terms:

```

IntegralError_next = IntegralError + PowerError * Delta_t_RRS
TiltIntegral_next = TiltIntegral + TiltError * Delta_t_RRS

ActuatorCommand = Bias
                  + K_P * PowerError
                  + K_I * IntegralError_next
                  + regional tilt terms

```

The controller does not directly add reactivity. It changes an actuator state, which a validated influence map converts into local absorption overlays before the next spatial solve.

8.2 Delays, rates, and event ordering

Requested command, available command, and physical actuator state are three different states. Commands have explicit delays, due times, bounds, deterministic IDs, and queue ownership. Physical motion is rate-limited and causal: a newly available command cannot retroactively affect the interval before its due time.

At an event time, the planned order is broadly:

```

burnup -> refuelling -> command generation -> actuator motion
        -> branch updates -> spatial solve -> kinetics/Xe step

```

Every state transition is explicit, ordered, versioned, and rollback-safe.

8.3 Influence maps

An influence map converts a control state into local absorption change:

```

DeltaSigma_a,g,i = sum_q Weight_(q,i,g)
                  * (ActuatorState_q - ReferenceState_q)

```

The map is sparse, versioned, and bound to an explicit node and group. It is not also added as a separate direct reactivity term. A sign certificate checks that an automatic controller acts as negative feedback.

8.4 Included branch schemas and deferred physics

The Phase 6 Core implementation contains synthetic/test-only contracts for:

- 14 logical liquid-zone regions grouped into 6 physical assemblies;
- adjuster banks with explicit insertion fractions and rate limits;
- bulk poison mass, concentration, and add/withdraw rates;
- controller projections, command queues, delays, saturation, and deterministic scenario evidence.

The four items above are bounded P6-T01 through P6-T07 contract evidence and are conditionally passed by G6. They are not a validated production controller, external-reference comparison, or golden RRS authority. The remaining branch schemas are still deferred:

- temperature states for fuel, coolant, or moderator; and
- moderator purity mass fraction.

Disabled temperature, purity, zone, adjuster, and poison branches contribute exactly zero. V1 has no thermal equation, CFD, chemistry law, coolant-flow model, or hidden cleanup law. Temperature and purity are prescribed branch states until a later task authorizes a dynamic model.

9. Numerical method summary

Problem	Planned method	Why it is useful for the game
Spatial neutronics	Two-group finite-volume operator plus sequential source/power iteration	Produces a spatial flux and power shape from explicit topology
Inner group solve	Backend-neutral deterministic linear solve with scalar residual acceptance	Allows a validated implementation without hiding convergence failure
Power scaling	Total fission-power normalization to caller target	Separates shape/criticality from the scenario's power scale
Burnup	Left-endpoint piecewise-constant power integration	Makes power history directly change future fuel state
Coefficient lookup	Strict-knot linear interpolation, no extrapolation	Makes burnup-dependent physics auditable and bounded
Point kinetics	Explicit Euler amplitude and delayed precursors	Adds time-dependent power response without full transient CFD
I/Xe	Explicit Euler production, decay, and absorption inventories	Makes recent power history affect future reactivity
RRS motion	Delayed commands plus rate-limited state updates	Turns control actions into understandable time-lagged choices
Feedback	Sparse local absorption overlays	Changes spatial shape and regional tilt rather than only total power
Determinism	Canonical ordering, finite double, explicit time, no unordered reductions	Makes saves, replays, tests, and player outcomes repeatable

The most important numerical design principle is **fail closed**. A failed inner solve, nonconverged outer solve, invalid coefficient, negative inventory, stale snapshot, or inconsistent event must produce a diagnostic and no usable partial state. The game can turn a failed scenario into a clear error or loss; it must not silently continue with invented physics.

For explicit Euler, the caller supplies a stability policy. The permitted substep is bounded by kinetic, nuclide, and actuator limits plus declared rate envelopes. If a scheduled gap is too long, it is subdivided deterministically; the simulator does not hide instability by choosing an undocumented step.

10. How this physics feeds the game

10.1 The player-facing loop

The planned game loop is:

1. **Inspect:** see channel and bundle power, burnup, residence history, regional tilt, xenon state, and available control actions.
2. **Choose:** refuel a legal channel, wait, change a regulating action, add or withdraw poison, or preserve margin for a later event.
3. **Advance:** the explicit scheduler processes burnup, transitions, delayed commands, spatial solves, kinetics, and isotope history.
4. **Explain:** report what changed and why, such as a local absorption change, a power-shape shift, a burnup increment, or a xenon penalty.
5. **Evaluate:** score survival, energy production, refuelling efficiency, stability, and avoidable control use.

10.2 Why each physics layer matters

Physics layer	Player consequence
Spatial flux and power shape	A channel can be locally hot or cold even when total power looks acceptable
k and reactivity	The same refuelling action can be stabilizing in one state and destabilizing in another
Burnup and coefficient interpolation	Old and fresh bundles are not interchangeable; history matters
Refuelling direction and atomicity	A legal move changes a whole train and has a visible discharge consequence
RRS and tilt	The player must balance total power against regional shape
Kinetics	Power does not jump arbitrarily; delayed neutrons create memory
Xenon	Recent high power can create a later absorption penalty and change future choices
Deterministic events and replay	The player can learn from consequences and reproduce a strategy

The game is successful when these layers make decisions meaningful without forcing the player to read raw solver internals. The user interface should show cause and effect in terms such as `power moved toward the north region or xenon absorption increased after sustained power` while retaining detailed diagnostics for developers.

11. What remains to be investigated

The following are not yet settled runtime facts:

- exact DRAGON5/DONJON5 build identity, nuclear-data checksums, coefficient generation, and lawful reference-data admission;
- the performance and evidence limits of the current scalar Jacobi inner solve;
- the production coefficient data pack, leakage conductances, and boundary conductances;
- direct production/full-core reference authority and production thresholds;
- an applicable RRS comparison package, evidence-backed G6 tolerances, and an unconditional G6 disposition;
- evidence-backed G7A and optional G7B tolerances and golden cases;
- the small synthetic topology and data fixtures used before full-core data;
- the controller influence-map values, sign certificate, and regional map;
- stable timestep policies for representative rate/decay envelopes;
- whether a later kinetics integrator is worth investigating against the v1 explicit-Euler baseline; and
- whether temperature and moderator-purity feedback improves the game enough to justify Phase 7B.

The literature digest is useful for methodology and candidate case design. It does not fill these gaps. In particular, values such as a reported burnup, shift size, group library, or controller arrangement remain candidate evidence until the complete case-admission proof is available.

12. Explicitly out of scope

This project does not implement or investigate as runtime behavior:

- shutdown, scram, trip, emergency, or safety-system response;
- accident progression or operator-training behavior;
- full thermal hydraulics, CFD, coolant chemistry, or detailed temperature evolution;
- OpenMC code, OpenMC-generated production data, or an OpenMC runtime;
- a named commercial station map or proprietary operating history;
- thorium, SCWR, accident-tolerant-cladding, or advanced-cycle runtime models;

- hidden random behavior, Unity-frame-driven physics, or nondeterministic parallel reductions.

13. Living-document maintenance and consistency checks

This guide is a derived teaching document. Repository rules govern execution; the implementation plan governs product/phase intent; the project scope register governs current status; specifications and ADRs govern technical decisions; code and task/gate reports provide implementation evidence. The P1-T08 literature digest supplies context and candidate-case evidence only.

Update the guide when any of the following changes:

- a physics specification, unit, sign, equation, state field, or numerical method changes;
- a runtime physics class or data-pack loader is implemented;
- a gate changes the status of implementation, tolerance, or golden evidence;
- the literature digest adds or reclassifies a relevant claim; or
- the game loop, scoring, player decision, or out-of-scope boundary changes.

Run the consistency check from the repository root:

```
& .\tools\Check-PhysicsGuideConsistency.ps1
```

Regenerate the PDF from this Markdown source:

```
python .\tools\build_physics_guide_pdf.py
```

The check verifies required headings, local links, current Phase 3/4/5/6, G4, G5, and G6 evidence, specification status markers, PDF freshness, and a readable PDF page count. It is intended to fail when the documented status and Core evidence diverge. PDF pages should still be visually inspected after major layout changes.

14. Traceability and references

Project documents

- Project scope and delivery register
- Implementation plan
- P2-T01 topology and units
- P2-T02 two-group solver
- P2-T03 refuelling and burnup
- P2-T04 kinetics, xenon, RRS, and feedback
- P2-T05 observables and validation
- P1-T08 task report
- CANDU literature digest v1
- G2 forced-closure addendum
- G3 state-contract review
- P4-T01 stencil assembly
- P4-T02 operator binding
- P4-T03 source/eigen iteration
- P4-T06-R2D reduced/synthetic admission disposition
- P4-T06-R3 reduced-model/interpolation boundary

- P4-T06-R4 candidate interpolation pack
- P4-T06-R5 candidate comparison snapshots
- P4-T06-G4G bounded mathematical admission
- G4-R3 blocked fresh gate
- P4-T06-G4K independent reduced-model authority
- G4-R6 reduced-model G4 disposition
- Reduced-model/interpolation boundary specification
- P5-T06 affected coefficient and spatial-state recomputation
- P5-T07 Phase 5 identity, location, burnup, and energy invariants
- P5-T08 deterministic multistep refuelling histories
- P5-T09 complete lifecycle and I/Xe contracts
- P5-T10 approved ReducedModel sequence comparison
- P5-T11 through P5-T16 Phase 5 correction chain
- G5 Phase 5 refuelling and depletion review
- P6-T07 bounded RRS scenario evidence
- G6 Phase 6 regulating-system review

Literature evidence boundary

The applicable P1-T08 rows for this guide are S1-R02, S1-R03, S1-R05, S1-R06, S1-R07, S1-R08, S1-R09, S1-R10, S5-R03, S5-R04, S5-R08, S6-R02, S6-R03, and S6-R05. They are used as MethodologySupport or CandidateCaseDesign context only. No row is used as an approved runtime constant or golden value.

The guide should be updated before a later task uses a literature row as candidate numeric evidence. That task must record the full case-admission proof: exact tool/build, nuclear-data identity, geometry, state/history, units, normalization, output definition, independent reproduction, and licensing.

15. Glossary

Term	First-year meaning
Absorption cross section	A measure of how likely a neutron is removed by absorption in a material
Burnup	Energy extracted per unit heavy-metal mass; here stored in J/kg_HM
Diffusion	A reduced model that estimates neutron movement from flux gradients
Eigenvalue	A value such as k that makes a balance equation self-consistent
Fission source	New neutrons produced by fission
Flux	Neutron flow intensity used to calculate reaction rates
Homogenization	Replacing detailed material regions by averaged group constants
Macroscopic cross section	A bulk interaction coefficient with units of inverse length
Precursor	A delayed-neutron-emitting fission product represented by a state variable
Reactivity	A signed measure derived from k that indicates a tendency to change power
Source iteration	Repeatedly solve a spatial balance using the latest fission source
Xenon poisoning	Absorption by Xe-135, whose concentration depends on recent power history

Change record

Version	Date	Change
2.8	2026-08-24	Refreshed the living handoff with direct Core 158/158 and Golden 19/19, Phase 6 focused 46/46, the TEST-INFRA-01 wrapper recovery, the completed Phase 6 refactor/naming chain, and the unchanged synthetic-only G6 limitation
2.7	2026-08-21	Added the G6 conditional synthetic/test-only disposition and recorded the then-current direct and artifact-wrapper validation boundary; kept external RRS authority deferred
2.6	2026-08-18	Added the ReducedModel channel-plane geometry visualization; preserved authority boundaries
2.5	2026-08-18	Added model cards, evidence mapping, and visualizations
2.4	2026-08-18	Recorded G4-R6 approval, the final G5 PASS for the bounded ReducedModel/Core scope, P5-T10 through P5-T16 evidence, and Phase 6/G6 activation; kept direct production/full-core authority deferred
2.3	2026-08-16	Recorded the G4-R3 BLOCKED disposition, the P4-T06-G4H next handoff, and the P5-T10/G5 dependency; kept production tolerances and golden status deferred
2.2	2026-08-16	Recorded the R2D-selected reduced/synthetic offline boundary, existing-table interpolation surface, direct Candu6.x2m admission retirement, and completed P5-T09 lifecycle/I-Xe contract status; kept R4/R5, G4/G5, reference comparisons, and golden consumers pending
2.1	2026-08-15	Recorded P5-T08 test-only deterministic multistep S4/S8 history evidence, explicit interval/refuelling ordering, shuffled-input replay equality, complete per-bundle/discharge trace coverage, and identity non-reuse; kept production history schema, reference comparisons, golden consumers, and G4/G5 approval pending
2.0	2026-08-15	Recorded P5-T07 explicit bundle-count, identity, location, nonnegative-burnup, exact energy-accounting, interval-energy, monotonicity, and refuelling partition invariants; kept complete lifecycle/I-Xe/power-history envelopes, deterministic long histories, reference comparisons, golden consumers, and G4/G5 approval pending
1.9	2026-08-15	Recorded P5-T06 explicit epoch/period/index cadence, burnup-selected node coefficient rebinding, preserved validated conductances, deterministic spatial solve, exact topology-instance binding, and converged atomic lifecycle acceptance; kept full scheduler/kinetics/I-Xe transitions, canonical checksum serialization, complete power-history binding, reference comparisons, golden consumers, and G4 approval pending
1.8	2026-08-15	Recorded P5-T05 immutable burnup-indexed coefficient values/table metadata, strict-knot validation, exact/linear lookup, derived <code>chi_2</code> , and fail-closed out-of-range behavior; kept canonical data-pack checksum verification, solver recomputation, complete lifecycle/history binding, reference comparisons, golden consumers, and G4 approval pending
1.7	2026-08-15	Recorded P5-T04 accepted bundle-power binding, explicit left-endpoint energy integration, derived burnup, immutable replacement, monotonicity/overflow validation, and proposed version advancement; kept full power-history/digest lifecycle binding, coefficient lookup, reference comparisons, golden consumers, and G4 approval pending
1.6	2026-08-15	Recorded P5-T03 named refuelling mapping/atomicity bodies, canonical position bindings, immutable paired event append, and bounded basic discharge audit records; kept complete I/Xe-bearing discharge/transaction envelopes, burnup, coefficient lookup, reference comparisons, golden consumers, and G4 approval pending
1.5	2026-08-15	Recorded P5-T02 immutable atomic directional movement over the current BundleInventory fields; kept complete lifecycle/I-Xe/event envelopes, burnup, coefficient lookup, reference comparisons, golden consumers, and G4 approval pending
1.4	2026-08-15	Recorded P5-T01 declarative S4/S8 scheme metadata and canonical flow-bound position-plan validation; kept atomic movement, burnup, reference comparisons, golden consumers, and G4 approval pending
1.3	2026-08-15	Recorded P4-T05 synthetic solved-case coverage; kept traceable reference comparisons, golden consumers, and G4 approval pending

Version	Date	Change
1.2	2026-08-15	Recorded P4-T04 outer convergence and fail-closed diagnostics as implemented; kept synthetic coverage, reference comparisons, golden consumers, and G4 approval pending
1.1	2026-08-15	Corrected implemented Phase 3/4 status; separated the one-iteration solver foundation from pending outer convergence, comparison, golden-data, and G4 work; aligned maintenance and authority references
1.0	2026-08-10	Initial extraction of implemented status, P2 physics contracts, planned numerical methods, game coupling, and maintenance checks